

CHIH-HAO (ANDY) TSAI

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Github Portfolio: <https://github.com/andytsai104/andytsai-robotics-portfolio>

Education

M.S. in Robotics and Autonomous Systems

Arizona State University, Tempe, Arizona, United States

Aug. 2024 – May 2026

GPA: 3.5/4.0

B.S. in Mechanical Engineering

National Taipei University of Technology, Taipei, Taiwan

Sep. 2019 – Jun. 2023

GPA: 3.03/4.0

Professional Experience

BELIV Lab, Arizona State University

Research Assistant

Jun. 2025 – Present

Mesa, Arizona

- Built a **BC+TD3 pedestrian controller** in CARLA for autonomous-vehicle safety testing.
- Developed BEV observations and continuous control for simulation-based system validation.
- Automated model training and testing on ASU SOL HPC using Bash scripts.

Test Research Inc.

AOI deployment Intern

Jul. 2022 – Sep. 2022

Taipei, Taiwan

- Assembled and supported the deployment of automated optical inspection machines.
- Assisted with basic equipment troubleshooting and operational checks during machine setup.

Publications

- **Chih-Hao Tsai**, H. Guo, and J. Zhao, “Closed-Loop Pedestrian Policy Learning for Safety-Critical Scenario Generation in Autonomous Driving,” *Modeling, Estimation and Control Conference (MECC)*, 2026 (Accepted).

Academic Projects

OmniVLA Semantic Warehouse Navigation

Jan. 2026 – Apr. 2026

- Integrated **OmniVLA-edge**, **ROS 2**, **Gazebo**, and **Nav2** into a semantic navigation pipeline.
- Built a ROS 2 data-collection pipeline for navigation testing and model fine-tuning.
- Tested goal prediction and autonomous navigation across simulated warehouse locations.

Consensus-Based Multi-Robot Warehouse Task Allocation

Aug. 2025 – Dec. 2025

- Built a **ROS 2** and **Gazebo** warehouse simulation with differential-drive robots and 2D LiDAR.
- Integrated task assignments with **Nav2** goal dispatch and robot-status monitoring.
- Troubleshot robot namespaces, localization, costmaps, and navigation configurations.

Pololu 3pi+ 2040 Embedded Robotics

Aug. 2025 – Dec. 2025

- Integrated IMU, encoders, IR sensors, bump sensors, and LCD telemetry on an embedded mobile robot.
- Troubleshot sensor readings, control logic, and robot motion using real-time telemetry.
- Tuned feedback control for line tracking, hill climbing, and obstacle avoidance.

Vision-Based Maze Solving & Path Planning with MyCobot Pro 600

Mar. 2025 – Apr. 2025

- Built a **ROS 2 control pipeline** for vision-based path execution on a 6-DOF robotic arm.
- Created a SolidWorks-based **URDF digital twin** and resolved robot kinematic issues in RViz and Gazebo.
- Validated planned trajectories on physical hardware through **TCP/IP control**.

Autonomous Mobile Vehicle and Robotic Arm

Feb. 2022 – Nov. 2022

- Built an autonomous mobile vehicle with an integrated robotic arm for object relocation.
- Integrated camera-based object detection, motor control, and robotic actuation on physical hardware.
- Designed and fabricated robot components using SolidWorks and 3D printing.

Technical Skills

- **Robotics & Integration:** ROS 2, Nav2, URDF/xacro, robotic arms, mobile robots, sensor integration
- **Testing & Validation:** equipment setup, operational checks, system testing, sim-to-real validation
- **Control & Embedded:** PID/PI control, odometry/IMU integration, real-time systems, motion control
- **Programming:** Python, C/C++, MATLAB, Bash
- **Engineering Tools:** Gazebo, RViz, Linux, Git, Simulink, SolidWorks, OpenCV, TCP/IP